BURHAN at Daleel Shared Task Subtask-1: A Multi-Label Transformer Pipeline for Arabic Argumentative Discourse Unit Classification

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Abstract

This paper presents our system for Task 1 of the Daleel 2026 shared task, which assigns multiple discourse-function labels to Arabic paragraphs from editorials and debate transcripts. We fine-tune a monolingual Arabic encoder (MARBERTv2) with a label-weighted focal loss, multilabel-stratified k-fold cross-validation, out-of-fold per-label threshold calibration, k-fold ensembling, and a genre-conditioning prefix, in a two-stage protocol that first trains on the training split alone and then retrains on training+development. Internal cross-validation macro-F1 rises from 0.667 to 0.680 across the two stages. On the official test set, the system attains micro-F1 = 0.766 and macro-F1 = 0.560, with markedly stronger micro-F1 on debate paragraphs (0.825) than editorials (0.633), and a smaller but consistent macro-F1 gap (0.554 vs. 0.456). *Assumption* is classified most reliably (F1 = 0.88–0.94) and *Common Ground* least reliably (F1 $\leq$ 0.12), mirroring the low inter-annotator agreement reported for the same category in the source English scheme. We find a $\sim$0.12 gap between internal cross-validation and official test macro-F1, concentrated in low-support, genre-sensitive categories. Our code is publicly available at `https://github.com/EidAhmed1/daleel2026_task1_project`.

1 Introduction

Argument mining for Arabic presents several challenges that arise from characteristics of the language itself, including rich morphology, variation in orthographic representation, inconsistent use of diacritics, and differences in writing conventions. These difficulties are further amplified by the comparatively limited availability of large-scale annotated Arabic argumentation resources when compared with English (Al-Khatib et al., 2016; Khader et al., 2024).

Existing Arabic resources are typically genre-level argumentative corpora without discourse-unit typing (Khader et al., 2024; Zaghouani et al., 2024), while prior modeling work has relied on machine-translated, projection-labeled English data, where projection noise is a dominant performance bottleneck (Nabhani and Al-Khatib, 2025). The Daleel 2026 shared task (Nabhani et al., 2026) contributes to addressing this limitation by introducing Arabic paragraphs from editorials and debates annotated according to their argumentative discourse-unit functions. In subtask 1, the classification problem is explicitly multi-label because an individual paragraph can perform several discourse functions simultaneously. For example, the same paragraph may state an assertion while also providing statistical evidence. Consequently, a successful system must identify all applicable discourse categories rather than selecting a single class.

Our submission makes four main contributions. First, we develop a multi-label learning pipeline that integrates multilabel-stratified k-fold cross-validation, out-of-fold (OOF) calibration of label-specific decision thresholds, and ensemble prediction across folds; stratification and per-label threshold calibration mitigate the effects of label imbalance, while fold ensembling primarily improves prediction stability. Second, we motivate our approach to class imbalance by relating it to findings on annotation difficulty reported for the closest comparable English argumentative discourse unit (ADU) annotation scheme. Third, rather than reporting only the post-retraining internal estimate, we report cross-validation results transparently at both training stages – train-only (used for development-set validation) and train+development (used for the final submitted model) – exposing any optimism in the internal estimate before it is compared against official test performance. Finally, we conduct genre-stratified evaluation and examine the discrepancy between cross-validation results and performance on the official test set. These components together provide

Proceedings of The Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026) - Volume 2 Shared Tasks, pages 1370–1374
October 28-29, 2026 ©2026 Association for Computational Linguistics

a systematic framework for training, calibration, evaluation, and analysis of the presented system.

The remainder of this paper describes the complete processing pipeline, beginning with data preparation and ending with final test prediction, while also explaining the rationale and practical considerations associated with the principal design decisions.

2 Background

Daleel Shared Task Subtask-1 is formulated as a multi-label classification problem over Arabic paragraphs, where each paragraph may be assigned one or more of six possible discourse-function labels: AS (Assumption), AN (Anecdote), ST (Statistics), TE (Testimony), CO (Common Ground), and OT (Other), forming the label set $L = \mathrm{AS}, \mathrm{AN}, \mathrm{ST}, \mathrm{TE}, \mathrm{CO}, \mathrm{OT}$. In addition to the discourse labels, each paragraph is associated with a genre attribute indicating whether it originates from an editorial or a debate. The training and development data contain 829 paragraphs in total, comprising 612 training and 217 development instances, while the test set contains 213 paragraphs (87 editorial, 126 debate), forming the official evaluation set described in Section 5.

The shared task scores per-label precision/recall/F1, aggregated into micro-, macro-, weighted-, and samples-averaged scores, over the combined set and separately per genre. Macro-F1 is the official task metric and treats all labels equally regardless of frequency; we additionally report micro-F1, which reflects overall label-assignment accuracy dominated by the most frequent labels, and weighted-F1, which weights each label's F1 by its support as a middle ground between the two. We additionally report leakage-free OOF cross-validation estimates during development, where each training example is scored only by fold models that excluded it.

The label inventory descends from Al-Khatib et al. (2016), who annotate 300 English editorials with substantial overall agreement (Fleiss' $\kappa = 0.56$) but much lower agreement for *Common Ground* ($\kappa = 0.114$) and *Other* ($\kappa = 0.152$), also the rarest categories (1.7% and 1.2% of units). We use this as external evidence that these categories are both rare and intrinsically hard, motivating our imbalance-aware training. Related but incompatible annotation schemes include the microtext scheme of Peldszus and Stede (2015) and the convincingness-oriented annotations of Habernal and Gurevych (2016).

On Arabic resources, Khader et al. (2024) and Zaghouani et al. (2024) release argumentative corpora (against general Arabic speech resources Al-Fetyani et al., 2023; Mubarak et al., 2021 and the bilingual corpus of Habash and Palfreyman, 2022) but without ADU-type labels compatible with our scheme. The closest modeling precedent, Nabhani and Al-Khatib (2025), compares cross-lingual transfer, translate-train, and zero-shot LLM prompting on the English essays of Stab and Gurevych (2017), machine-translated with NLLB (Costa-jussà et al., 2022) and label-projected via FastAlign (Dyer et al., 2013), following Eger et al. (2018). They find a monolingual Arabic encoder (Antoun et al., 2020) trained on translated data beats multilingual alternatives (Conneau et al., 2019) and zero-shot LLMs, but projection noise (64–69% paragraph-level error) dominates over translation error itself (0.6% token-level). Because our data is natively Arabic rather than translated and projected, this dominant noise source does not apply by construction, while we retain the preference for a monolingual encoder.

3 System Overview

3.1 Pipeline

The pipeline begins by parsing the JSON Lines files into a consistent tabular representation in which each paragraph is associated with its text, genre, and, when available, gold labels. The text is then normalized, with an optional genre-specific prefix added to provide explicit information about the source type. The gold labels are converted into fixed six-dimensional binary vectors corresponding to the target discourse categories.

In Stage 1, five multilabel-stratified folds are trained on the training split, and OOF sigmoid probabilities are collected for each label. These pooled OOF probabilities are subsequently used to determine a separate decision threshold for each label by maximizing binary F1. The five Stage-1 models are then ensembled by averaging their predictions and evaluated on the development set.

In Stage 2, the same training and threshold-calibration procedure is repeated after combining the training and development sets. Finally, the Stage-2 ensemble, together with the calibrated label-specific thresholds, is used to generate the predictions submitted for the test set.

3.2 Data processing

A regex normalizer strips diacritics and tatweel, unifies alef/alef-maksura/ya and hamza variants, collapses repeated-character runs and whitespace, applied identically to all splits. A short Arabic genre token ("opinion article:"/"debate:") is optionally prepended. Tokenization uses the model's tokenizer with 320-subword truncation and padding.

3.3 Model

We select MARBERTv2 over alternative Arabic encoders such as CAMeLBERT because it is pretrained on a large corpus spanning both Modern Standard Arabic and dialectal/social-media text, which we expect to transfer better to our mixed genres (formal-register editorials and spoken-register debates) than encoders pretrained predominantly on MSA news text; we do not ablate this choice against other encoders and leave a systematic comparison to future work.

We fine-tune `UBC-NLP/MARBERTv2` with a linear six-way sigmoid head (dropout 0.3), freezing the embeddings and first four encoder layers throughout both stages.

3.4 Imbalance handling

We use a label-weighted focal loss plus per-label threshold calibration, with a supplementary lexical rule boosting Statistics recall.

For Statistics, if the threshold is not reached but a percentage sign or a lexical marker (e.g., "study", "survey/poll") appears and $\hat{P}_c > 0.15$, the label is still assigned.

4 Experimental Setup

No hyperparameter optimization was conducted. Instead, all training configurations were predetermined and kept constant, as summarized in Table 1. Experiments were carried out on Google Colab, with an approximate runtime of 4–6 minutes per fold for Stage 1 and 5–10 minutes per fold for Stage 2. The implementation relied on Hugging Face `transformers`/PyTorch, along with scikit-learn and `iterative-stratification` for model development and data splitting.

Setting	Value
Encoder	MARBERTv2
Head	Linear, 6 outputs, sigmoid
Dropout	0.3
Frozen layers	Embeddings + first 4
Loss	Focal (γ=2.0), $\alpha \in [.5, .9]$
Optimizer / LR	AdamW / 2×10^{-5}
Weight decay / warmup	0.01 / 0.10
Epochs (max) / batch	8 / 16
Early stopping	Patience 3, F1@0.5
CV	MultilabelStratified, K=5
Max length	320 subwords

Table 1: Training configuration.

5 Results

5.1 Cross-Validation

Table 2 reports per-label OOF thresholds and F1. Stage 1 (train only, $N\approx612$) reaches macro-F1 = 0.6667; Stage 2 (train+dev, $N\approx829$) reaches 0.6799 (+0.0132). AS, ST, and TE improve with more data; AN, CO, and OT decline slightly, with CO showing the largest relative drop (−0.0667). The mean fixed-0.5-threshold macro-F1 across folds (0.6652 Stage 1, 0.6548 Stage 2) is consistently below the pooled, per-label-calibrated OOF macro-F1, confirming that threshold calibration helps over a fixed cutoff.

Label	Stage 1		Stage 2	
	t^*	F1	t^*	F1
AS	0.49	0.920	0.53	0.929
AN	0.51	0.715	0.47	0.708
ST	0.51	0.455	0.59	0.559
TE	0.53	0.734	0.57	0.795
CO	0.51	0.400	0.51	0.333
OT	0.45	0.778	0.51	0.756
Macro	–	0.667	–	0.680

Table 2: Per-label out-of-fold (OOF) decision thresholds (t^*) and binary F1, comparing Stage 1 (trained on the training split only, $N\approx612$) and Stage 2 (trained on training+development, $N\approx829$). The Macro row reports macro-averaged F1 across all six labels.

5.2 Official Evaluation

Table 3 reports overall metrics and Table 4 per-label F1 by evaluation condition (N=213 combined, 87 editorial, 126 debate). The combined micro-F1 is 0.766 and macro-F1 is 0.560. The performance is strongly genre-dependent: micro-F1 reaches 0.825 on debates but only 0.633 on editorials (a 19-point gap), with a smaller but consistent macro-F1 gap (0.554 vs. 0.456). The cross-validation macro-F1 (≈0.68) exceeds the official combined macro-F1 (0.560) by about 0.12 absolute.

5.3 Discussion

Threshold calibration consistently beats a fixed 0.5 cutoff. However, cross-validation overestimates

Set	Micro-F1	Macro-F1	Wtd-F1
Combined	0.766	0.560	0.760
Editorial	0.633	0.456	0.595
Debate	0.825	0.554	0.828

Table 3: Official overall metrics by evaluation condition.

Label	Comb.	Edit.	Debate
AS	0.920	0.881	0.941
AN	0.551	0.492	0.625
ST	0.333	0.429	0.000
TE	0.620	0.270	0.790
CO	0.095	0.000	0.118
OT	0.840	0.667	0.849
Macro	0.560	0.456	0.554

Table 4: Official per-label F1 by evaluation condition.

test generalization by $\sim$0.12 macro-F1, concentrated in low-support categories (CO, ST) whose small size inflates CV variance. Editorial paragraphs are consistently harder than debate paragraphs, suggesting the single genre-prefix mechanism under-models register differences; the Statistics rule in particular fails to generalize to debate. Early stopping in most Stage-1 folds also suggests a mismatch between the fixed-threshold model-selection criterion and the per-label thresholds used at inference.

6 Conclusion

We present a multi-label Arabic ADU classification system for Daleel 2026 shared task subtask-1, combining a monolingual encoder, focal loss, stratified k-fold CV, OOF threshold calibration, ensembling, and two-stage training. The system attains official micro-F1 = 0.766 and macro-F1 = 0.560, with *Assumption* and *Other* classified most reliably and *Common Ground* least reliably, mirroring low human agreement on that category in related English annotation. Internal cross-validation overestimates official test macro-F1 by $\sim$0.12, a gap concentrated in rare, genre-sensitive labels. Future work includes ablating the loss, genre-conditioning mechanism, and Statistics rule; genre-specific calibration; and confusion-matrix-level error analysis once per-example predictions are available.

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